

Analysis First-Year Exam, January 2018, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $(a_n : n \geq 0)$ be a bounded real sequence. For each $N \geq 0$ define $\bar{a}_N = \sup\{a_n : n \geq N\}$ and let $s = \inf\{\bar{a}_N : N \geq 0\}$. Prove:
 - (i) if $s < t$ then $a_n < t$ eventually, in the sense $(\exists N)(\forall n \geq N)(a_n < t)$;
 - (ii) if $s > r$ then $a_n > r$ frequently, in the sense $(\forall N)(\exists n \geq N)(a_n > r)$.
2. Let X be a metric space. Assume that there exists $\delta > 0$ such that the distance between distinct points of X is always δ or greater. Determine precisely which subsets of X are open, which closed, which compact and which connected.
3. Let $f : X \rightarrow \mathbb{R}$ have the property that if a is any real number then $U(a) = \{x : f(x) < a\}$ is an open subset of the compact metric space X .
 - (i) Prove that f is bounded above on X .
 - (ii) Prove that $s = \sup\{f(x) : x \in X\}$ is a value of f . For (ii): suppose s not a value and consider the function $F = 1/(s - f)$.
4. Let $f : X \rightarrow Y$ be a bijection between metric spaces; assume that Y is complete and that f^{-1} is continuous.
 - (i) Show that if f is uniformly continuous then X is complete.
 - (ii) Show that if f is only continuous then X can fail to be complete.
5. Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable and assume that its derivative f' is bounded. Prove that the right-hand limit $\lim_{t \rightarrow a+} f(t)$ exists.